This assignment will not be graded and is only for practice.

Level 1

1) The dimension of a vector space is the : ***1 point***

- [ ] Cardinality of any spanning set.
- [ ] Cardinality of a basis.
- [ ] Cardinality of minimal spanning set.
- [ ] Cardinality of maximal linearly independent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Cardinality of a basis.
Cardinality of minimal spanning set.
Cardinality of maximal linearly independent set.

2) Match the sets of vectors in column A with their properties of linear dependence or independence in column B and the dimension of the vector spaces in column C spanned by the sets. ***1 point***

	Set of vectors (Column A)		Linear dependence or independence (Column B)		Dimension of the vector space spanned by the set (Column C)
a)	$\{(1,0,0),(0,1,0),(1,1,1),(1,0,1)\}$	i)	Linearly independent	1)	1
b)	$\{(1,0,-1),(0,-1,0),(-1,1,-1)\}$	ii)	Linearly dependent	2)	2
c)	$\{(2,3,6),(\frac{1}{3},\frac{1}{2},1),(4,6,12)\}$	iii)	Linearly dependent	3)	3
d)	$\{(1,0,-\frac{1}{6}),(0,1,-\frac{1}{3}),(3,2,-\frac{7}{6})\}$	iv)	Linearly dependent	4)	3

Table : M2W4AQ3

- [ ] a → ii → 3, b → iii → 2
- [ ] c → iii → 2, d → iv → 1
- [ ] a → ii → 4, b → i → 3
- [ ] c → iv → 1, d → iii → 2

No, the answer is incorrect.
Score: 0

Accepted Answers:

a → ii → 4, b → i → 3
c → iv → 1, d → iii → 2

3) Choose the set of correct options. ***1 point***

- [ ] The dimension of the vector space $M_{1\times 2}(\mathrm{R})$ is 2.
- [ ] The dimension of the vector space $M_{2\times 1}(\mathrm{R})$ is 1.
- [ ] The dimension of the vector space $M_{3\times 3}(\mathrm{R})$ is 3.
- [ ] A basis of $M_{2\times 2}(\mathrm{R})$ is the set $\{\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 1\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\1 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}\}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

The dimension of the vector space $M_{1\times 2}(\mathrm{R})$ is 2.
A basis of $M_{2\times 2}(\mathrm{R})$ is the set $\{\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 1\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\1 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}\}$

4) Consider the following sets: ***1 point***

$V_1 = \{A \mid A \in M_{2\times 2}(\mathrm{R}) \text{ and } A \text{ is a symmetric matrix, i.e., } A = A^T\}$
$V_2 = \{A \mid A \in M_{2\times 2}(\mathrm{R}) \text{ and } A \text{ is a scalar matrix}\}$
$V_3 = \{A \mid A \in M_{2\times 2}(\mathrm{R}) \text{ and } A \text{ is a diagonal matrix}\}$

$V_4 = \{A \mid A \in M_{2\times2}(\mathbb{R})$ and A is an upper triangular matrix$\}$
$V_5 = \{A \mid A \in M_{2\times2}(\mathbb{R})$ and A is a lower triangular matrix$\}$
All $V_i, i = 1, 2, 3, 4, 5$ are subspaces of the vector space $M_{2\times2}(\mathbb{R})$.
Choose the set of correct options.

- [ ] The dimension of V_1 is 3.
- [ ] The dimension of V_2 is 3.
- [ ] The dimension of V_3 is 1.
- [ ] The dimension of V_4 is 3.
- [ ] The dimension of V_5 is 3.

No, the answer is incorrect.
Score: 0

Accepted Answers:

The dimension of V_1 is 3.
The dimension of V_4 is 3.
The dimension of V_5 is 3.

5) Consider the following two matrices M_1 and M_2. ***1 point***

$$M_1 = \begin{bmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{bmatrix}, M_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Choose the correct set of options.

- [ ] The row rank of both the matrices, M_1 and M_2, is 2.
- [ ] The row rank of both the matrices, M_1 and M_2, is 3.
- [ ] The column rank of M_1 is 4, but the column rank of M_2 is 3.
- [ ] The column rank of M_2 is 4, but the column rank of M_1 is 3.
- [ ] The column rank of both the matrices, M_1 and M_2, is 4.
- [ ] The column rank of both the matrices, M_1 and M_2, is 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The row rank of both the matrices, M_1 and M_2, is 3.
The column rank of both the matrices, M_1 and M_2, is 3.

Level 2

6) Find the dimension of the vector space $V = \{A \mid$ sum of entries in each row is 0, and $A \in M_{3\times 2}(\mathbb{R})\}$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

7) Find the dimension of the vector space $V = \{(x, y, z, w) \mid x + y = z + w, x + w = y + z,$ and $x, y, z, w \in \mathbb{R}\}$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

8) Choose the set of correct options. ***1 point***

- [ ] If $S \subset \mathbb{R}^2$ contains three vectors, then vectors in S must be linearly independent.
- [ ] If $S \subset \mathbb{R}^3$ contains three vectors, then vectors in S may or may not be linearly independent.
- [ ] If $S \subset \mathbb{R}^4$ contains five vectors, then vectors in S must be linearly dependent.
- [ ] If $S \subset \mathbb{R}$ contains one vector, then the vector in S must be linearly dependent.

No, the answer is incorrect.
Score: 0

Accepted Answers:

If $S \subset \mathbb{R}^3$ contains three vectors, then vectors in S may or may not be linearly independent.

If $S \subset \mathbb{R}^4$ contains five vectors, then vectors in S must be linearly dependent.

9) Find the dimension of the vector space $V = \{A \in M_{3\times3}(\mathbb{R}) : A = A^T\}$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 6

1 point

10) Find the dimension of the vector space $V = \{A \in M_{3\times3}(\mathbb{R}) : A = -A^T\}$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Your score is: 0/10